

Rongfeng Cheng | Medical AI Research

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Education

South China University of Technology (SCUT)

Guangzhou, China

Bachelor of Management in Big Data Management and Application

2023–2027

Cumulative GPA: 3.46/4.0; average score: 85.18/100 (through Semester 6).

Selected AI coursework: Large Models and Generative AI (93/100), Large Language Models and Prompt Engineering (93/100), Machine Learning (92/100), and Deep Learning (88/100).

Selected Achievements & Deployed Work

Published: Co-authored a study on ethical safety in LLM code generation, published in *iScience* (2026).

IP: Two first-inventor patent applications and two software copyrights.

Clinical Deployment: Co-developed Dingbei Xinxin, deployed at Guangdong Second Provincial General Hospital; led the five-person Dingbei Meimei engineering and clinical-workflow team through its hospital launch.

Awards: Campus Gold, Internet+ Competition; Second Prize, Guangzhou Mathematical Modeling University League; Third Prize, CUMCM Guangdong Division.

Research Experience

Institute of Healthcare Artificial Intelligence Application, Guangdong Second Provincial General Hospital

DLEF: Explaining Why ECG Models Fail Across Hospitals

Jun 2026–Present

First author; submitted to *IEEE BIBM 2026*

- **Problem:** Aggregate performance scores show that a model degrades at a new hospital, but do not identify which diagnostic labels or evidence distributions drive the failure.
- **Contribution:** Formulated the question and led study design, implementation, experiments, analysis, and writing; developed directional dataset-label “evidence fingerprints.”
- **Finding:** Greater fingerprint mismatch predicted larger external AUROC degradation (Spearman $r = 0.843$ across 108 comparisons; $r = 0.892$ after aggregation), enabling label-level transfer-risk screening.

Matched-Control Perturbation Auditing: Does an ECG Model Use Clinical Evidence?

Jan 2026–Present

First author; submitted to *IEEE JBHI*

- **Problem:** Accurate models may rely on spurious cues, while target-only perturbations can confuse generic signal damage with genuine use of clinical evidence.
- **Contribution:** Developed and led a matched-control audit covering clinically defined evidence units, perturbation operators, and frozen-weight external validation.
- **Finding:** Only 6 of 15 model-diagnosis pairs were evidence-supported; matched controls or operator choice changed several conclusions, showing that strong external performance alone does not establish clinically intended evidence use.

Internship & Product Leadership

AI Research Institute, Guangdong Second Provincial General Hospital

Nov 2024–Present

LLM Development Engineer Intern

- Developed a standardized-patient Q&A pipeline from approximately 8,000 cardiovascular case PDFs, covering LLM-based information extraction, entity disambiguation, Neo4j knowledge representation, and hybrid retrieval.
- Explored knowledge-graph paths as structural scaffolds for generating medical question–answer and reasoning-trace data for model fine-tuning, evaluation, and retrieval-augmented generation.
- Led full-stack and clinical-workflow development for the five-person Dingbei Meimei acne and facial-health assessment team; supported its initial release when the hospital’s Refractory Acne Diagnosis and Treatment Center opened in July 2026.

- Designed a UMLS-grounded schema and normalization pipeline for heterogeneous health-examination data extracted from multi-institutional reports, including LLM-assisted schema evolution and terminology mapping.
- Developed backend workflow logic for the Health Navigation Plan within the company's iTwins product.

Products & Team Projects

Dingbei Xinxin / CardioGPT: Cardiovascular Proactive Health LLM

2025–Present

Co-founder & Core Developer; Development Lead since Aug 2026

- Co-developed a cardiovascular health-management platform integrating PPG/ECG data, intelligent consultation, risk assessment, and proactive health workflows.
- Led model fine-tuning, RAG, and full-stack implementation using LoRA, DPO, MQ-RAG, and chain-of-thought methods; also prepared grant and competition application materials.
- Deployed the initial mini-program at Guangdong Second Provincial General Hospital and currently leads full-stack development of version 2.0; supported by national-level innovation-program projects in 2025 and 2026 and incorporated as Yujian Future Mind Technology (Zhongshan) Co., Ltd. in Aug 2026.

Myocardial Ischemia Knowledge Graph

2025

Project Lead, SCUT Student Research Program

Advisor: Xunde Dong. Led a five-student interdisciplinary team; designed the ontology, annotation schema, guidelines, and quality-control workflow; built a multi-annotator platform with adjudication and agreement checks; and implemented XGBoost-based implicit reasoning with graph-mapped interpretability analysis.

Publications & Manuscripts

Published [1]: J. He, S. Qin, H. Zou, et al., **R. Cheng**, H. Zhang, and W. Cheng. "CodeSafetyBench: Evaluating and Improving Ethical Safety in Large Language Model Code Generation." *iScience*, 2026. doi:10.1016/j.isci.2026.115073.

Revised [2]: "Implicit Medical Ethical Deficits in Large Language Models: A Benchmarking and Alignment Study." *Nature Communications*; led the revision-stage experiments for the response to reviewers.

Under Review [3]: **R. Cheng et al.** "DLEF: Dataset-Label Evidence Fingerprints for Explaining Cross-Dataset Performance in ECG Classification." *IEEE BIBM 2026*. First author.

Under Review [4]: **R. Cheng et al.** "A Matched-Control Perturbation Audit Framework for Evaluating Evidence Use in Deep Learning Models: An Application to 12-Lead ECG Classification." *IEEE JBHI*. First author.

Ongoing: PathCLIP: Evidence-Aware ECG–Text Representation Learning. ECG–text alignment and zero-shot recognition with matched-count, permuted-hierarchy, residual-only, and held-out-label controls.

Skills

Programming & Frameworks: Python, PyTorch, scikit-learn, TypeScript, SQL

Modeling Methods: XGBoost, representation learning, multimodal learning, LLM fine-tuning, optimization

LLM & Knowledge Systems: LoRA, DPO, RAG/MQ-RAG, knowledge graphs, information extraction, UMLS

Evaluation & Data Curation: RAG/retrieval evaluation, perturbation analysis, external validation, schema design

Engineering: Neo4j, Next.js, Flask, Git, Docker

Languages

Chinese (native) Cantonese (native) English: IELTS 6.5, CET-6 528